New York University **Spring 2026**
Department of Economics

Econometrics II (Panel Data)

Graded Problem Set April 2026

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This problem set consists of three exercises covering (1) efficiency comparisons in linear static models, (2) income processes via dynamic models and (3) testing for correlated random coefficients. Each exercise asks you to answer some theory questions and code up simulations or empirical calculations. You must try to solve all three exercises in full.

When asked to do asymptotic approximations, feel free to assume LLNs and CLTs apply (i.e., you do not need to specify regularity conditions) but explain your derivations in detail.

Grading will be done according to the criteria explained in the syllabus to the course.

Due: April 28, 2026. Please submit your solutions to both instructor and TA by email.

Exercise 1: First Differences vs Within Groups

Consider the following static linear panel data model:

$$y_{it} = \beta x_{it} + \eta_i + v_{it}, \qquad i = 1, \ldots, N, \quad t = 1, \ldots, T,$$

with strict exogeneity with respect to an instrument z_{it},

$$E[\, v_{it} \mid \eta_i, z_{i1}, \ldots, z_{iT} \,] = 0.$$

We adopt a fixed effects perspective meaning that we leave $E[\, \eta_i \mid z_{i1}, \ldots, z_{iT} \,]$ unrestricted. Let $v_i = (v_{i1}, \ldots, v_{iT})'$ and $z_i = (z_{i1}, \ldots, z_{iT})'$, and denote $\Sigma(z_i) = \mathrm{Var}(\, v_i \mid z_i \,)$. We will parametrize $\Sigma(z_i)$ as the covariance matrix of the process:

$$v_{it} = \rho v_{i,t-1} + \varepsilon_{it}, \qquad t = 2, \ldots, T,$$

with $\mathrm{Var}(\, v_{i1} \mid z_i \,) = \sigma^2$ and $\mathrm{Var}(\, \varepsilon_{it} \mid z_i \,) = \sigma^2$.

The regressand y_{it} and endogenous regressor x_{it} are scalar whereas the instrument z_{it} is of dimension $r \geq 1$. The goal of this exercise is to compare estimators that arise when transforming the data using first differences and within groups.

(a) Write down the formula for the 2SLS estimators that use (i) Δz_{it} as an instrument in the first-difference equation (FD) and (ii) $z_{it} - \bar{z}_i$ (bars denote individual means) as an instrument in the within-group equation (WG):

$$\Delta y_{it} = \beta \Delta x_{it} + \Delta v_{it}, \tag{FD}$$

$$y_{it} - \bar{y}_i = \beta(x_{it} - \bar{x}_i) + v_{it} - \bar{v}_i. \tag{WG}$$

Call them $\hat{\beta}_{\text{FD}}$ and $\hat{\beta}_{\text{WG}}$. Obtain their asymptotic distributions as $N \to \infty$ with T fixed.

(b) Assume linearity of the first stage in levels: that is, for some $r \times 1$ full-rank matrix Π,

$$E[\, x_{it} \mid z_{i1}, \ldots, z_{iT} \,] = \Pi' z_{it}.$$

Obtain the (unfeasible) optimal IV estimator for β under the linear first-stage assumption and our model of $\Sigma(z_i)$, and its asymptotic variance. Both the unfeasible estimator and the asymptotic variance will depend on Π, ρ and σ^2. For which values of the parameters (if any) are $\hat{\beta}_{\text{FD}}$ or $\hat{\beta}_{\text{WG}}$ as efficient as the optimal IV estimator?

(c) Run a Monte Carlo simulation to compare $\hat{\beta}_{\text{FD}}$ and $\hat{\beta}_{\text{WG}}$ with the following guidelines:

- Simulate $n_{\text{MC}} = 10000$ samples with $N = 2000$ units and $T = 7$ time periods. Make units i.i.d. in the cross-section and set $\beta = 1$.
- Set $r = 2$, $\Pi = (1, 1)'$ and $\sigma^2 = 1$. Run simulations for $\rho \in \{-0.2, 0, 0.2, \ldots, 1\}$.
- Draw z_i from a normal distribution with mean and variance of your choice. Then, to generate endogeneity, draw $x_{it} = \Pi' z_{it} + \alpha \varepsilon_{it} + \sqrt{1 - \alpha^2} u_{it}$, where u_{it} is independent of v_i and z_i, and serially independent $N(0, \sigma^2)$. Set $\alpha = 0.4$.
- In each sample, compute $\hat{\beta}_{\text{FD}}$, $\hat{\beta}_{\text{WG}}$ and the (unfeasible) optimal IV estimator $\hat{\beta}_{\text{IV}}$.
- Generate a figure with ρ in the horizontal axis and the Monte Carlo estimates of the sampling variances of $\hat{\beta}_{\text{FD}}$, $\hat{\beta}_{\text{WG}}$ and $\hat{\beta}_{\text{IV}}$ (scaled by N) in the vertical axis.
- Experiment by changing some of the other parameters (Π, σ^2, α) to values of your choice. Generate no more than 3 extra figures.

(d) In your words, what is the main lesson from this exercise? In a given empirical situation, what prior information about the data generating process would lead you to choose the first-difference estimator over the within-groups estimator?

Exercise 2: Income and Consumption

In the file `ps1_data.csv` you will find panel data on income, consumption, wealth and age for $N = 792$ households for $T = 6$ biennial waves (1999-2009) from the PSID.[1] The goal of this exercise is to use covariance structures to separate permanent and transitory income shocks, and to identify consumption responses to them as in the Hall-Mishkin framework.[2]

We will model observed income as

$$y_{it} = \eta_{it} + \varepsilon_{it},$$

where the persistent component follows a random walk,

$$\eta_{it} = \eta_{i,t-1} + u_{it},$$

and the transitory component follows an MA(1),

$$\varepsilon_{it} = v_{it} + \psi_\varepsilon v_{i,t-1}.$$

The errors u_{it}, v_{it} are serially and uncorrelated and mutually orthogonal at all lags and leads.

We will model changes in consumption as

$$\Delta c_{it} = \alpha u_{it} + \alpha\beta v_{it} + \left(e_{it} + \psi_c e_{i,t-1}\right),$$

where the first part captures responses to income shocks and the second is an MA(1) model for measurement error where e_{it} is serially uncorrelated and orthogonal to u_{it}, v_{it} at all lags and leads. Assume constant variances: $\text{Var}(\, u_{it}\,) = \sigma_\eta^2$, $\text{Var}(\, v_{it}\,) = \sigma_\varepsilon^2$ and $\text{Var}(\, e_{it}\,) = \sigma_c^2$.

This model specifies the covariance structure for $s_i \equiv (\Delta y_{i2}, \Delta c_{i2}, \ldots, \Delta y_{iT}, \Delta c_{iT})'$ as a

[1] These data correspond to the panel used in the paper Arellano, Blundell, Bonhomme (2017): "Earnings and Consumption Dynamics: A Nonlinear Panel Data Framework," *Econometrica*, 85(3), 693-734.

[2] The reference for this exercise is Hall, Mishkin (1982): "The Sensitivity of Consumption to Transitory Income: Estimates from Panel Data on Households," *Econometrica*, 50(2), 461-481. Note that we will use a simplified version of the Hall-Mishkin model but we are also working on biennial data.

function of the parameters

$$\theta = (\sigma_\eta^2, \psi_\varepsilon, \sigma_\varepsilon^2, \alpha, \beta, \psi_c, \sigma_c).$$

We are interested in the relative proportion of persistent vs transitory shocks $\sigma_\eta^2/\sigma_\varepsilon^2$ and the coefficients that govern the response of consumption α and β.

(a) Obtain an expression for $\Omega(\theta) = \mathrm{Var}(\, s_i \,)$ linking parameters to the covariance structure.

(b) Explain how you would exploit that link to estimate the parameters.

(c) Produce estimates and standard errors of the parameters of the income process $(\sigma_\eta^2, \psi_\varepsilon, \sigma_\varepsilon^2)$ using only income data. What do we learn about $\sigma_\eta^2/\sigma_\varepsilon^2$? Interpret.

Compare (heuristically; no formal test) data and estimated covariance matrices for income growth. Do you see any deviation? If so, what changes to the model specification may improve fit (richer dynamics for η_{it} or for ε_{it})?

(d) Produce estimates and standard errors of all θ using the full covariance structure. What can we conclude about α, β and the permanent income hypothesis?

Exercise 3: Testing for Correlated Random Coefficients

Consider the random coefficients model with strictly exogenous regressors:

$$y_{it} = \alpha_i + \beta_i x_{it} + \gamma' z_{it} + v_{it}, \qquad E[\, v_{it} \mid \beta_i, x_{i1}, z_{i1}, \ldots, x_{iT}, z_{iT} \,] = 0,$$

for units $i = 1, \ldots, N$ and periods $t = 1, \ldots, T$ where $T \geq 3$.

Denote $w_{it} = (x_{it}, z_{it}')'$. The dimensions are $k_w = \dim(w_{it}) = 1 + k$ where $k = \dim(z_{it})$. The goal of this exercise is to explore the testability of random effects-type assumptions on $E[\, \beta_i \mid w_i \,]$ where $w_i = (w_{i1}, \ldots, w_{iT})'$.

(a) Suppose a researcher ignores the heterogeneity in the slope β_i and decides to estimate a regression in first differences to deal with the presence of α_i. More specifically,

$$\begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \left(\sum_{i=1}^{N} \sum_{t=1}^{T} \Delta w_{it} \Delta w_{it}' \right)^{-1} \sum_{i=1}^{N} \sum_{t=1}^{T} \Delta w_{it} \Delta y_{it}'.$$

Assuming $E[\, \beta_i \mid w_i \,]$ unrestricted, obtain the probability limits of $\hat{\beta}$ and $\hat{\gamma}$ as $N \to \infty$ with T fixed and compare them to $\bar{\beta} = E[\, \beta_i \,]$ and γ. What happens if $E[\, \beta_i \mid w_i \,] = \bar{\beta}$?

(b) There is a function $g(w_i)$ such that $\mathrm{Cov}(\,\beta_i, g(w_i)\,) = 0_{(k+1)\times 1}$ implies $\hat{\beta} \xrightarrow{\mathrm{p}} \bar{\beta}$ and $\hat{\gamma} \xrightarrow{\mathrm{p}} \gamma$. Determine that function using your answer to part (a).

Next, consider the following model for the conditional expectation of the random slope:

$$E[\,\beta_i \mid w_i\,] = \lambda_0 + \lambda_1' g(w_i).$$

Derive a test of the null hypothesis

$$H_0 : \lambda_1 = 0_{(k+1)\times 1}$$

and describe how to implement it in detail.

(c) Using the same dataset as for exercise 2 consider applying this model with y_{it} = income, $x_{it} = t$ and z_{it} = age squared. Implement the test you derived in (b) and explain whether there is evidence of correlation between the heterogeneous slope and age squared.

Report estimates and standard errors of the mean slope following the approach supported by the result of the test, and compare them to the procedures discussed in class.